There is a global need to end dependence on coal or 'dirty energy' to facilitate a move toward a cleaner future. The idea is to phase out the use of coal for the generation of electricity by 2030 for developed countries and 2040 for developing countries. This will make sure that no new coal plants for power generation. This can enable countries to honour the Powering Past Coal Alliance (PPCA), which was formed to phase out the use of coal in the generation of electricity. However, with the ever-growing need for electricity, the move appears feeble.

Coal-fired power plants still supply most of the electricity

FOSSIL FUEL FIGHTBACK

The burning of fossil fuels for the generation of electricity has been the norm now for several years. Fossil fuel power plants work on coal or oil to generate heat which drives turbines, which produce electricity. This process of energy creation is responsible for carbon emissions. However, the demand for the growing power sector is met by these power plants. The effort to replace such a set system will take years to come. The growing need for electricity, to this date, is more or less completely met by the burning of fossil fuels. Even though renewable sector is growing at a commendable rate, it is not enough to match the growing demand for electricity.

To put things into perspective the growth of demand for electricity will grow by 100 GW globally in 2022. Over the same period, renewable energy capacity will grow by 35 GW. This difference in the demand must be met by the traditional burning of fossil fuels which brings us back to the problem of carbon emissions. IEA reports that global demand for electricity grew by almost 5 per cent in 2021, more than half of which was supplied by fossil fuels. The report further states that renewable energy capacity grew by 8 per cent in 2021 and is estimated to grow by 6 per cent in 2022. Even with such an increase, renewable sources might only be able to meet half or even less of the rising global demand in the same years. Fossil fuel will even then account for 40-45 per cent of the increased electricity demand in 2021-22.

To meet the renewable energy targets, fossil fuel generated electricity has to decline by 6 per cent a year. However, this is set to hit an all-time high this year, even in terms of carbon dioxide emission. This puts the clean global electricity prospect in a dilemma. Replenishable sources also face other problems that fossil fuels do not. Governments' building capacity of wind farms, solar panels and hydro projects is one thing, but the creation of electricity to meet the estimated demand is completely a different ball game.

There might be another problem of availability of the natural resources when required. Weather change cannot be predicted with 100 per cent accuracy. The unavailability of wind or ample sunlight is highly possible. Countries need to think in terms of total output to be generated at the user end. Renewable energies can under deliver at certain periods of time. For instance, during September there is not enough available wind power over Europe and China. This can lead to power outages in the cleaner future the world has planned for. To meet demand at such a time, power sectors need to rely again on coal, oil and natural gas.

Using fossil fuels as a replacement for renewable energy is a better alternative than the other way round. Especiialy in such cases. One can be conservative by building higher capacity renewable energy generators to cover more and phase out coal dependability. Till coal as an energy source has been completely wiped off the table, carbon capture, essentially can keep targeting the high emitting plants to keep a check on the emissions.

Carbon capture and storage is the process by which carbon dioxide released from heavy power plants, such as coal fired power plants, is captured before it enters the atmosphere. CO^2 can be absorbed by machinery and stored (carbon sequestration) for centuries. This can be an effective way to reduce emissions till the time renewable sources of energy arrive at a larger capacity.

Major companies are also cutting down their investments in 'dirty' sources of energy. Investment in 2021 was about $350 billion which was well below the levels seen previously in 2019. This can, in the long run, result in a problem for the future as economies are not spending enough and there is

A Carbon Capture and Storage Plant